



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

DIGITAL FORENSICS

System Forensics, Investigation, and Response

Lesson 7 Email Forensics

Learning Objective

- Summarize various types of digital forensics.

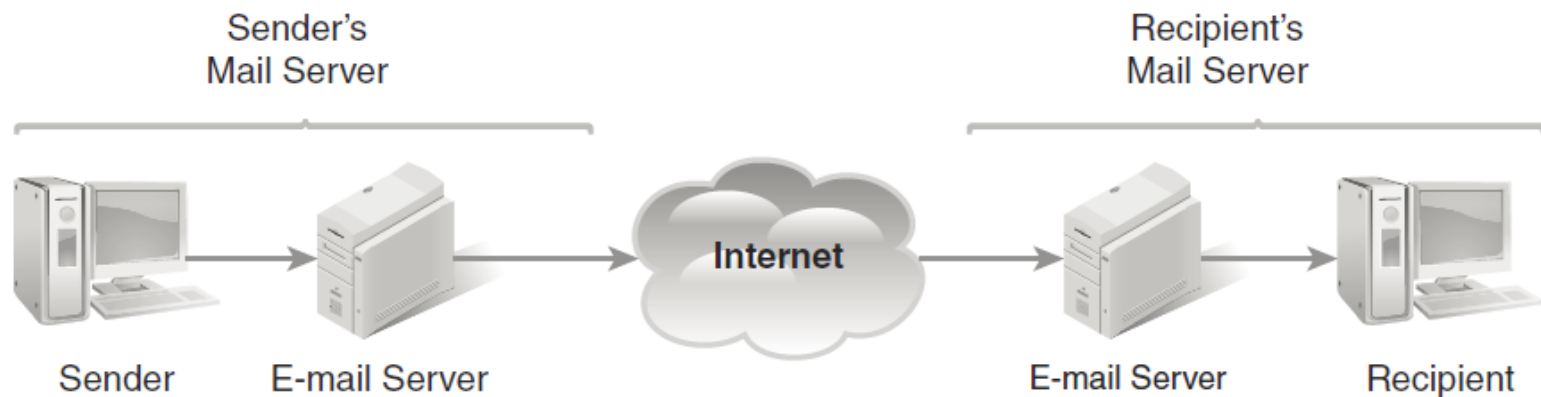
Key Concepts

- Email clients and servers
- Email headers
- Email tracing
- Email server forensic examination
- Laws related to email investigations

How Email Works

1. Sender uses a mail client to send a message
2. Message travels to multiple mail servers
 - Each mail server sends the message closer to its destination
3. Destination mail server stores the message
4. Receiver uses a mail client to retrieve the message from mail server

How Email Works



What an Email Review Can Reveal

- Email messages related to the investigation
- Email addresses related to the investigation
- Sender and recipient information
- Information about those copied on the email
- Content of the communications
- Internet Protocol (IP) addresses



What an Email Review Can Reveal (Cont.)

- Date and time information
- User information
- Attachments
- Passwords
- Application logs that show evidence of spoofing



Email Protocols

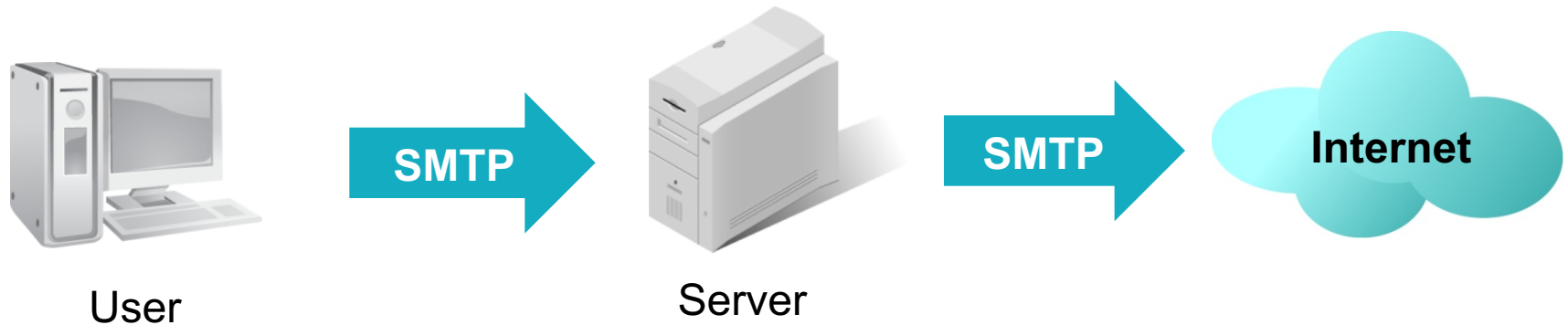
- Simple Mail Transfer Protocol (SMTP)
 - Used to send email from a client to a mail server, and between servers
 - Typically operates on port 25
 - SMTPS (secure) operates on port 465

Email Protocols (Cont.)

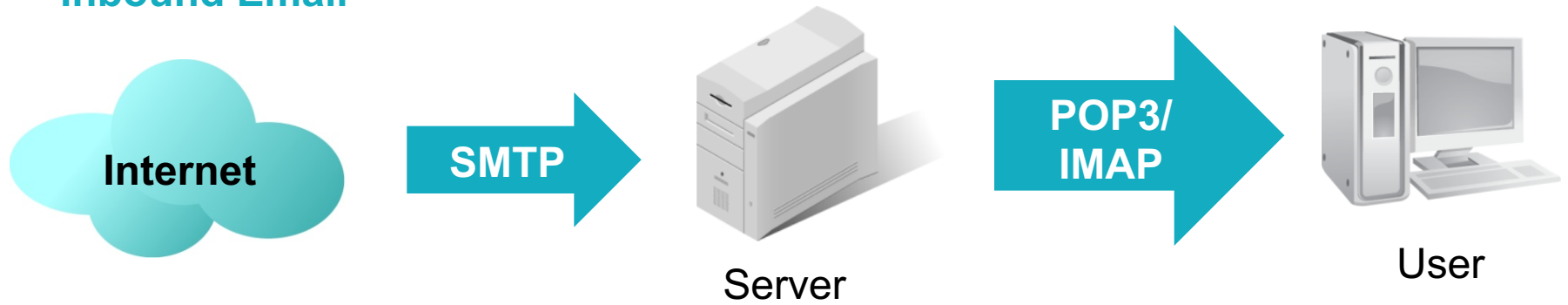
- Post Office Protocol version 3 (POP3)
 - Used to receive email
 - Operates on port 110, or 995 (secure)
 - Designed to delete email on server as soon as user downloads email
- Internet Message Access Protocol (IMAP)
 - Used to receive email
 - Operates on port 143
 - User views email on the server, decides whether to download the mail; email is retained on server

Email Protocol Process

Outbound Email



Inbound Email



Faking Emails

Spoofing

Anonymous
remailing

"Valid"
emails

Spoofing

- Making an email message appear to come from someone or someplace other than the real sender or location
- First machine to receive spoofed message records machine's real IP address
- Header contains both the faked IP and the real IP address

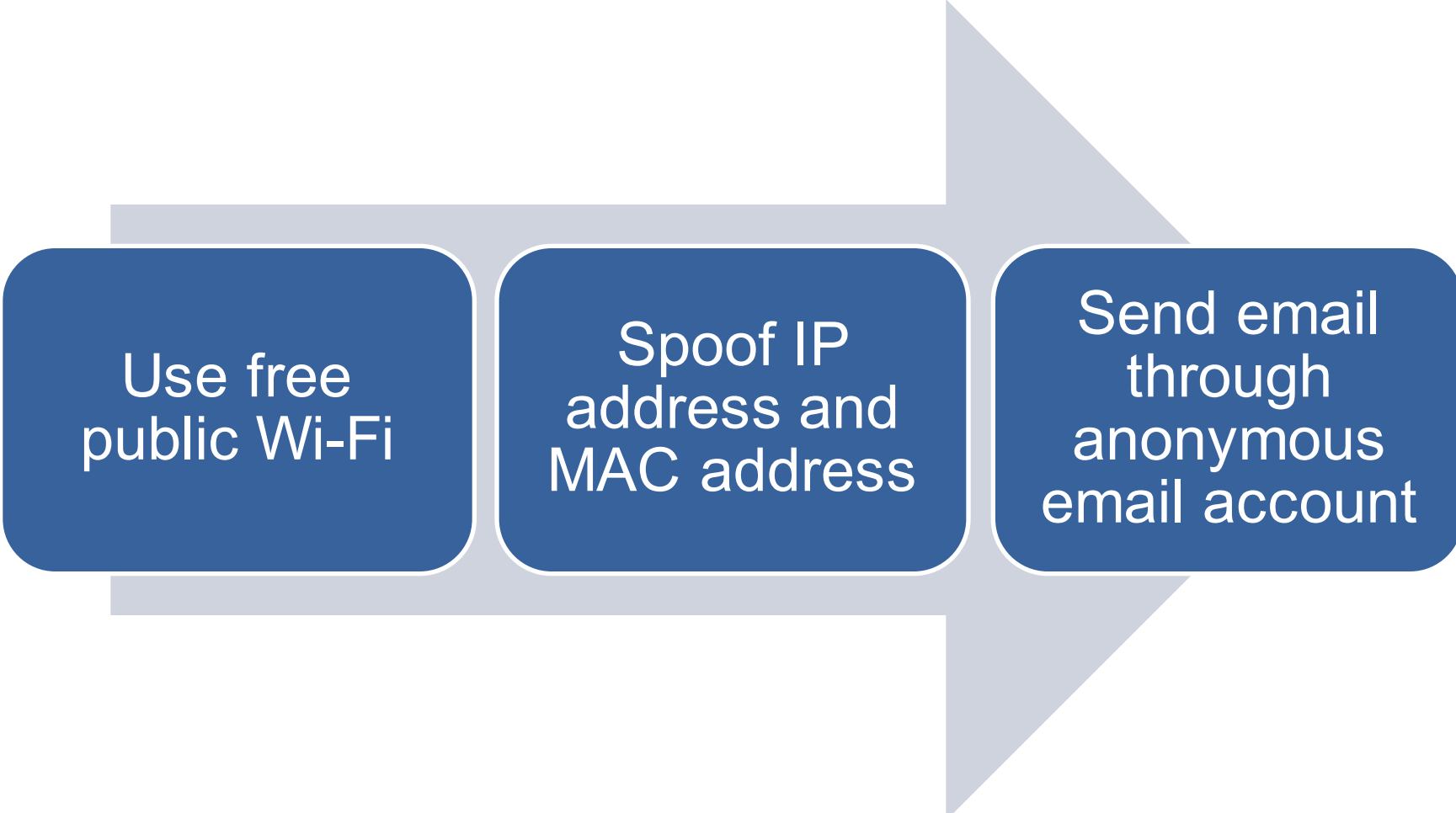
Anonymous Remailing

- Suspect sends an email message to an anonymizer
 - Anonymizer is email server that strips identifying information from message before forwarding it with anonymous mailing computer's IP address
- To find out who sent remailed email, must examine logs maintained by remailer or anonymizer companies

"Valid" Emails

- Appears as though mail is from trusted source
- Message content is suspicious
- Content may contain URL that points to malicious site

How to Fake an Email



Use free
public Wi-Fi

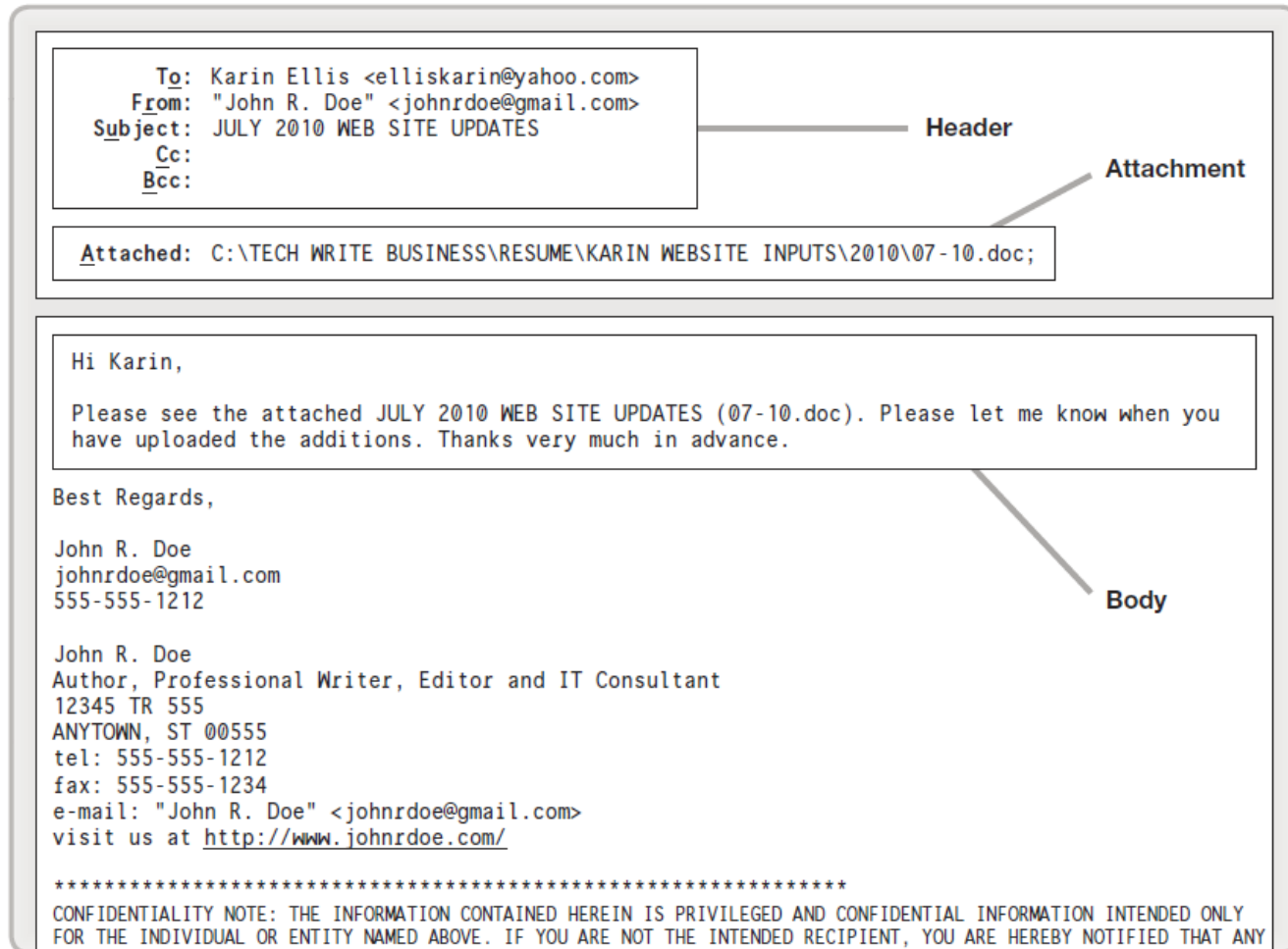
Spoof IP
address and
MAC address

Send email
through
anonymous
email account

Email Message Components

- Header
 - Addressing information
 - Source and destination
- Body
 - Contents of the message
- Attachments
 - External data that travels along with each message

Email Message Components



Email Headers

- RFC 2822
 - Standard for email format, including headers
- All email programs use the same email format, regardless of operating system
 - Email from Outlook on a Windows 10 PC can be read by recipient using Hotmail on Android phone

Email Headers (Cont.)

- Header keeps record of the message's journey networks and mail servers
- Each server adds information to the header
- Each network device has an Internet Protocol (IP) address
 - Identifies device
 - Can be resolved to a location address



RFC 2822 Specifications for Email Headers

Message header **must** include:

From field

The email address and, optionally, the name of the sender

Date field

The local time and date when the message was written

RFC 2822 Specifications for Email Headers (Cont.)

Message header **should** include:

Message-ID field
An automatically generated field

In-Reply-To field
The message-ID of the message that this is a reply to

Email Header Fields (RFC 3864)

To

Subject

Cc/Bcc

Content-
Type

Precedence

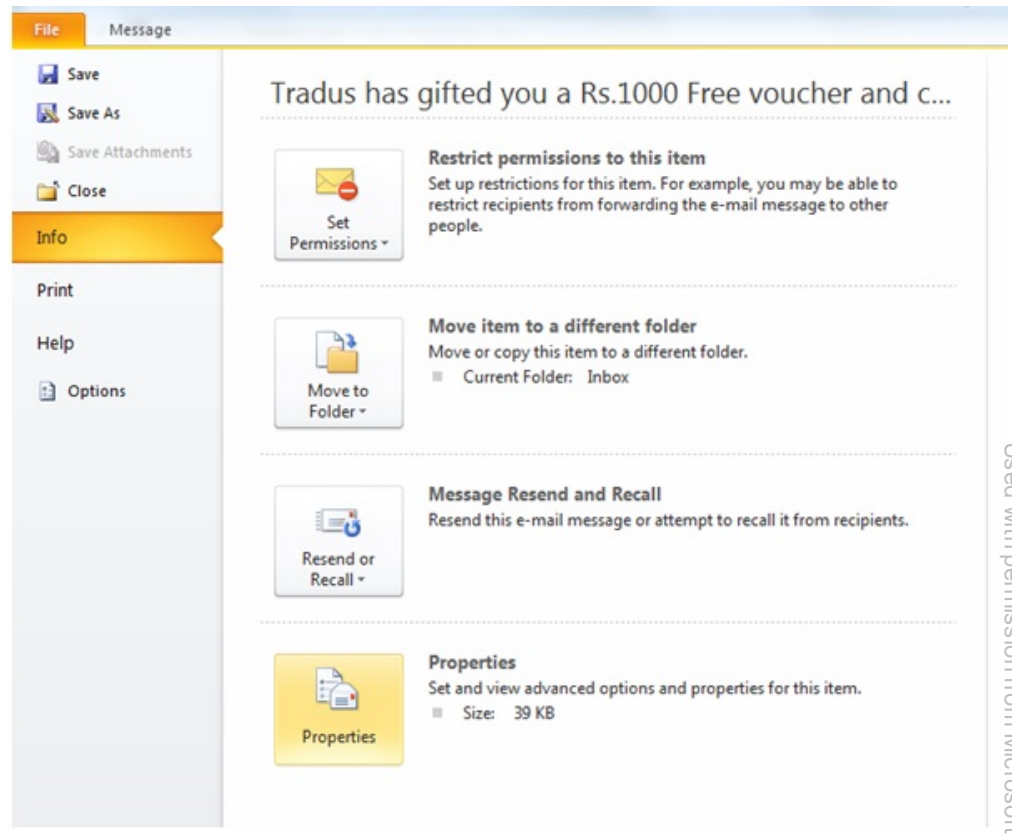
Received

References

Reply-To

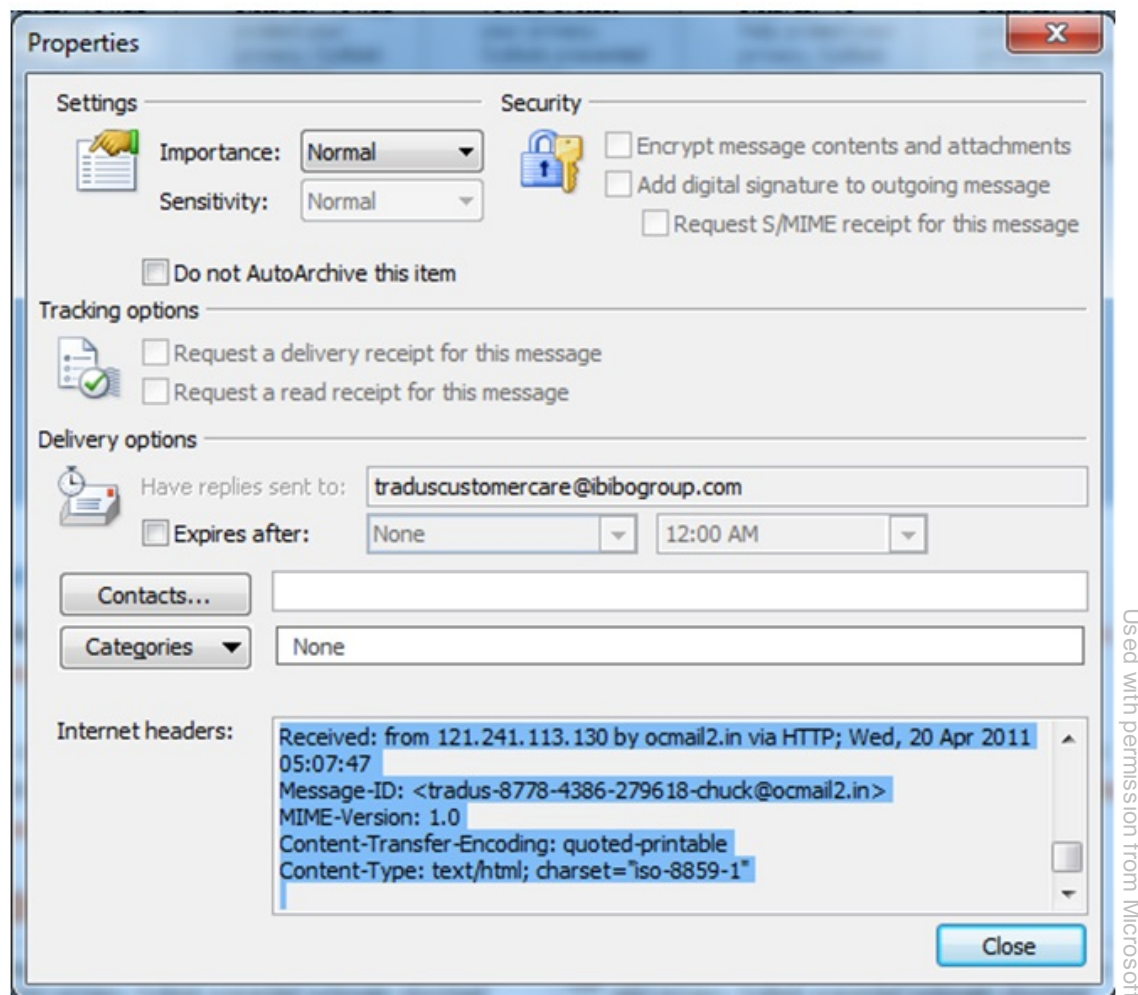
Sender

Find Microsoft Outlook 2010 Headers



Step 1

View Outlook 2010 Headers



Step 2

Microsoft Outlook 2010 Headers

Properties

Settings

Importance: Normal

Sensitivity: Normal

Do not AutoArchive this item

Tracking options

Request a delivery receipt for this message

Request a read receipt for this message

Delivery options

Have replies sent to:

Expires after: None 12:00 AM

Contacts...

Categories: None

Internet headers:

X-Apparently-To: chuck@chuckeasttom.com via 98.138.83.188; Wed, 10 Apr 2013 08:04:35 -0700
Received-SPF: none (domain of dcamedicalconsulting.com does not designate permitted sender hosts)
X-YMailISG: 2bYM4nQWLDuIsJ36V5UOgd5D8BVYTFp08Ycza4kkuVQVBcOJ
tLm_vg8W_hI0MmKjZMUakvW8HrpO2ItGpoQ1uTmKgZFOTshuzHc7Qdy9cMp

Close

Used with permission from Microsoft

Find Yahoo! Headers



The image shows a screenshot of a web interface. At the top is a solid blue horizontal bar. Below it is a white rectangular area containing a dropdown menu with the text "Select Message Encoding" and a downward-pointing chevron icon. To the right of the dropdown menu is a vertical line, followed by the text "[Full Headers](#)".

Courtesy of Yahoo!

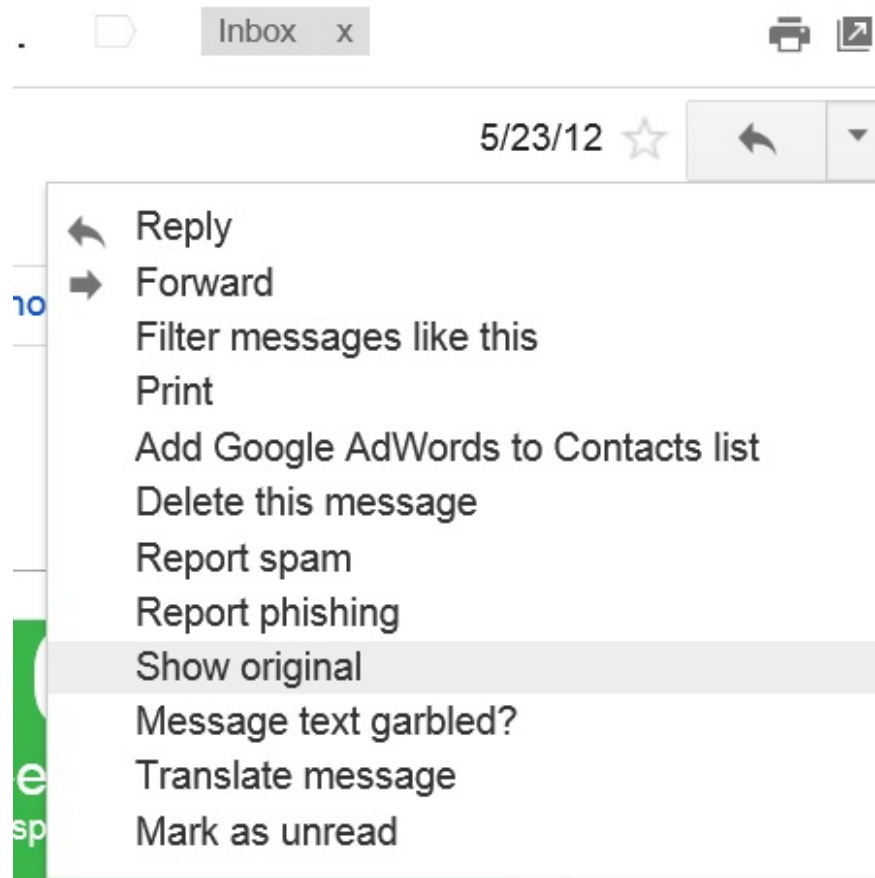
View Yahoo! Headers

amazon.com Author Central Team Wed Dec 16 02:17:03 2009

X-Apparently-To: chuckedsttom@yahoo.com via 206.190.48.191; Tue, 15 Dec 2009 18:17:09 -0800
Return-Path: <20091216021703f0099751069046bfa5eaa966e3b48697@bounces.amazon.com>
X-YMailISG: s6zas7sWLDuS2kDU9iAL1VpupHZ4KVrj1cdgEupzMKO.MrIXY67vKqIzvUPVOPigkLI5troH.hSSkKAGCHYD2YnNHSgLC
X-Originating-IP: [207.171.164.35]
Authentication-Results: mta1098.mail.re4.yahoo.com from=amazon.com; domainkeys=pass (ok); from=amazon.com; dkim=neutral (no s
Received: from 127.0.0.1 (EHLO mm-notify-out-1101.amazon.com) (207.171.164.35) by mta1098.mail.re4.yahoo.com with
DomainKey-Signature: s=rte02; d=amazon.com; c=nofws; q=dns; h=Date:From:To:Message-ID:Subject:MIME-Version: Content-Type:C
Date: Wed, 16 Dec 2009 02:17:03 +0000 (UTC)
From:  "Amazon.com Author Central Team" <ac-no-reply@amazon.com> 
To: "chuckedsttom@yahoo.com" <chuckedsttom@yahoo.com>
Message-ID: <1882985069.21977901260929823826.JavaMail.correios@na-mm-relay.amazon.com>
Subject: Amazon.com Author Central Update - December 15, 2009
MIME-Version: 1.0
Content-Type: text/plain; charset=UTF-8
Content-Transfer-Encoding: quoted-printable
AMAZON-CLIENT-HOST: cscentral-batchportal-processor-na-6002.iad6.amazon.com
Bounces-to: 20091216021703f0099751069046bfa5eaa966e3b48697@bounces.amazon.com
AMAZON-CLIENT-SENDBTIME: Wed Dec 16 02:17:03 UTC 2009
AMAZON-MAIL-RELAY-TYPE: notification
AMAZON-RTE-VERSION: 2.0
Content-Length: 3178

Courtesy of Yahoo!

Find Gmail Headers



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View Gmail Headers

Delivered-To: chuckedstom007@gmail.com
Received: by 10.182.25.137 with SMTP id c9csp84759obg;
Wed, 23 May 2012 13:58:23 -0700 (PDT)
Return-Path: <3b0-9Tw8KD4Ykn6yln2-xylozv8qyyqvo.mywmr4muok233ywAAHqwksw.myw@scoutcamp.bounces.google.com>
Received-SPF: pass (google.com: domain of 3b0-9Tw8KD4Ykn6yln2-xylozv8qyyqvo.mywmr4muok233ywAAHqwksw.myw@scoutcamp.bounces.google.com designates 10.50.40
Authentication-Results: mr.google.com; spf=pass (google.com: domain of 3b0-9Tw8KD4Ykn6yln2-xylozv8qyyqvo.mywmr4muok233ywAAHqwksw.myw@scoutcamp.bounces.g
Received: from mr.google.com ([10.50.40.230])
by 10.50.40.230 with SMTP id a6mr20555688igl.2.1337806703082 (num_hops = 1);
Wed, 23 May 2012 13:58:23 -0700 (PDT)
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed;
d=google.com; s=20120113;
h=date:from:to:cc:subject:reply-to:x-google-ads-sender:message-id
:x-trak-extra-language:mime-version:content-type;
bh=XekTgcSliYSkFAHQTAJcm7aGSzICw8DGRTeLhcUe8M8=;
b=H6rb7OD+xtuyUTC1NCDwh+e3HlJphFiN1BO431WP2SkZuYqn3BT/2XJ2muJslF+7A0
g716xr4Q/a+pquw0lyOQyMCFRQVBZFxFxThD90f6HdYHje/9GiMdu/YS+LrDbYhlWmgKqJ
nV9CMQyAj1lcOFkAqAOcd9EIDqJ8elnTGan8U+r18OUrKtL83R4rauiXIObnpdBrDF3t
ykwzmC42LXMEls53FCjM+Abn/geyxEB4wKmAfmvpxBbOOUC8S1bzs6PgqURKj112V1Xb
rcvDOazwpdqhlLPRND8a9PpFv2RHzWtPnl3thYQdljmKB1/vEivVV9XMHGbelj9StGy6
1PNA==
Date: Wed, 23 May 2012 20:58:23 +0000
Received: by 10.50.40.230 with SMTP id a6mr11939917igl.2.1337806703051;
Wed, 23 May 2012 13:58:23 -0700 (PDT)
From: Google AdWords <adwords-noreply@google.com>
To: chuckedstom007@gmail.com
Cc:
Subject: =?utf-8?q?Only_days_left_to_get_=_24100_=_26_tips_from_Google_AdWords=2E_En?=
=?utf-8?q?ds_May_31=2E?=
Reply-To: adwords-noreply@google.com
X-Google-Ads-Sender: auto_adsense_emails
Message-ID: <29060e2c.1337806701.000000.53518.1.EN_US.2f59020b@google.com>
X-Trak-Extra-Language: EN_US
MIME-Version: 1.0
Content-Type: multipart/alternative; boundary="=====0825779531=="

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View Hotmail Email Headers

1. Select Inbox from the menu on the left.
2. Right-click the message for which you want to view headers, and select View Message Source.

The full headers will appear in a new window.

View Apple Mail Email Headers

1. Open Apple Mail.
2. Click on the message for which you want to view headers.
3. Go to the View menu.
4. Select Message, then Long Headers.

The full headers will appear in the window below your Inbox.

Email Files

.pst
(Outlook)

.ost
(Offline Outlook
Storage)

.mbx or .dbx
(Outlook Express)

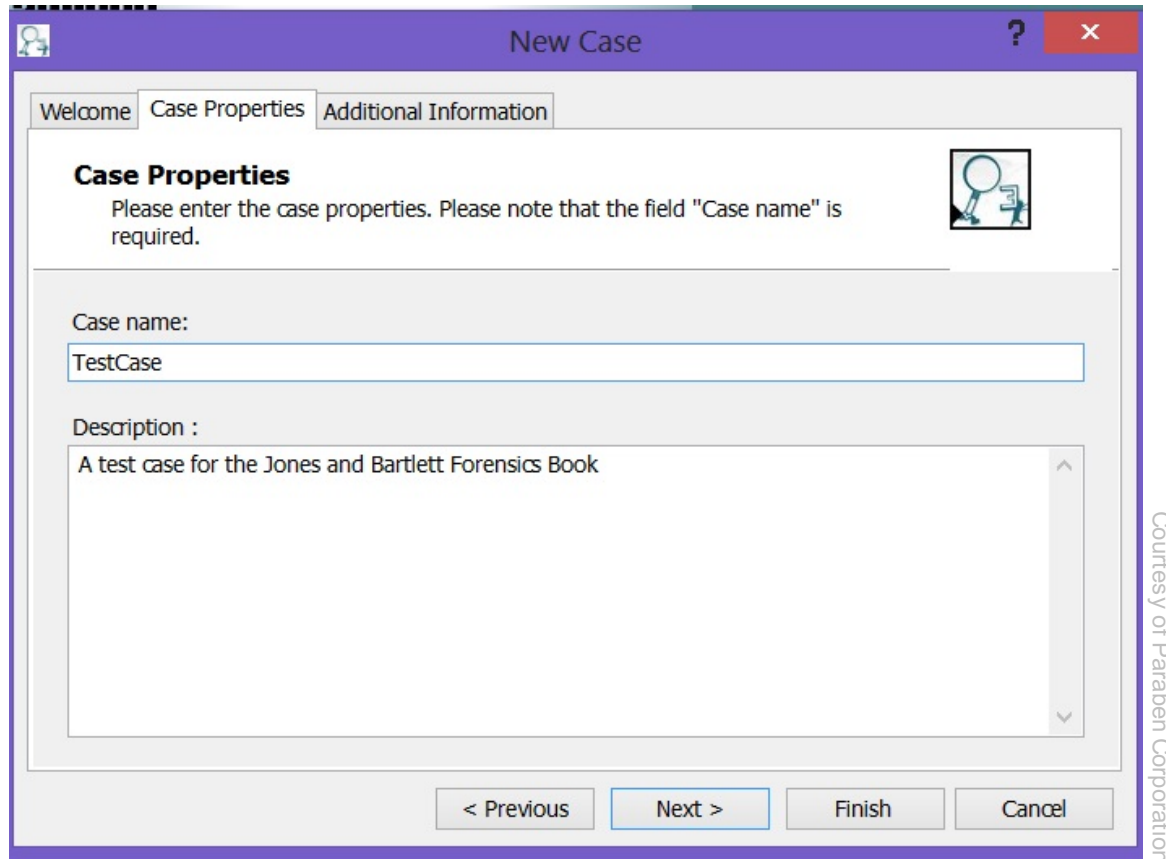
.mbx
(Eudora)

.emi
(common to
several email
clients)

Paraben's Email Examiner

- Exclusively for email forensics
- Works like the more complete forensic suites (Forensic Toolkit and EnCase) in that evidence is grouped by case

Creating a Paraben Case



The screenshot shows a 'New Case' dialog box with three tabs: 'Welcome', 'Case Properties', and 'Additional Information'. The 'Case Properties' tab is active. It contains a title 'Case Properties' and a note: 'Please enter the case properties. Please note that the field "Case name" is required.' Below this, there is a 'Case name:' label and a text input field containing 'TestCase'. Underneath is a 'Description :' label and a text area containing 'A test case for the Jones and Bartlett Forensics Book'. At the bottom, there are four buttons: '< Previous', 'Next >', 'Finish', and 'Cancel'. A vertical text label 'Courtesy of Paraben Corporation' is positioned to the right of the dialog box.

New Case

Welcome Case Properties Additional Information

Case Properties

Please enter the case properties. Please note that the field "Case name" is required.

Case name:

TestCase

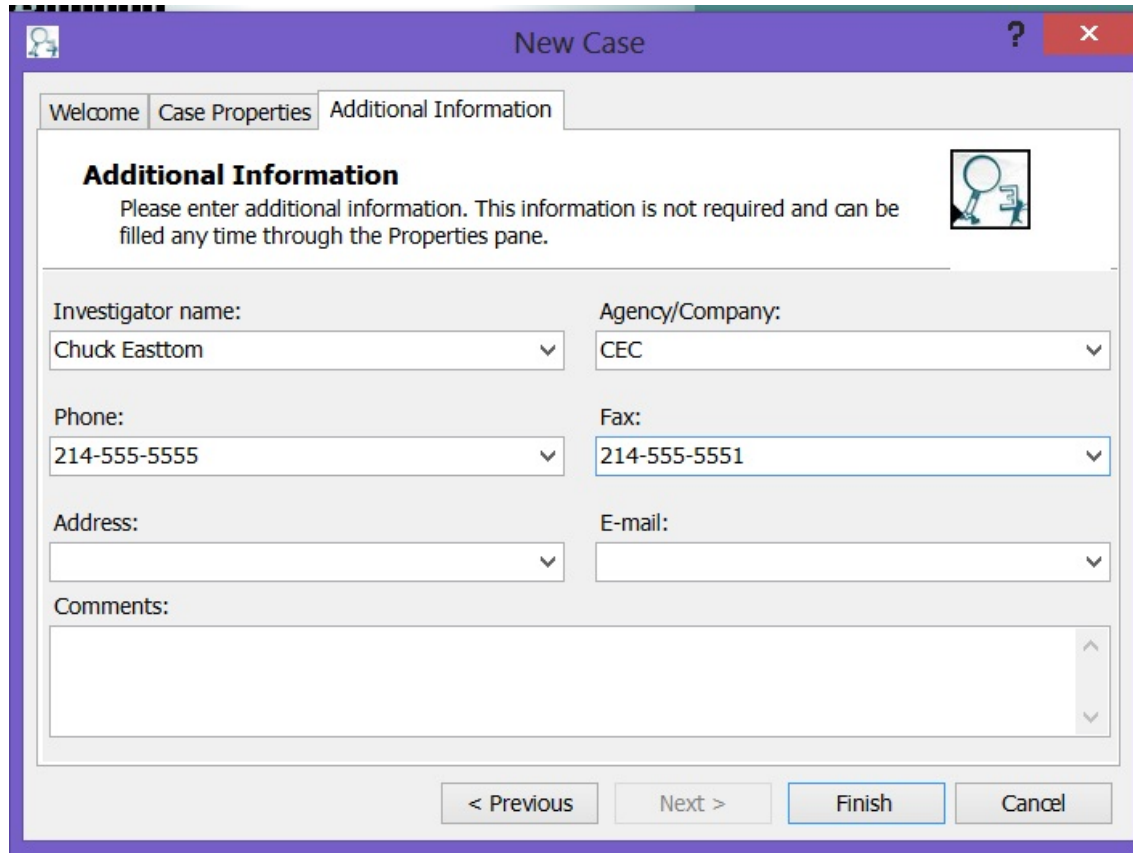
Description :

A test case for the Jones and Bartlett Forensics Book

< Previous Next > Finish Cancel

Courtesy of Paraben Corporation

Adding the Investigator



The screenshot shows a software window titled "New Case" with a purple header bar. It contains three tabs: "Welcome", "Case Properties", and "Additional Information". The "Additional Information" tab is active, displaying a magnifying glass icon and the text: "Additional Information. Please enter additional information. This information is not required and can be filled any time through the Properties pane." Below this, there are several input fields arranged in two columns. The left column contains "Investigator name:" (dropdown menu with "Chuck Easttom"), "Phone:" (text box with "214-555-5555"), "Address:" (dropdown menu), and "Comments:" (text area). The right column contains "Agency/Company:" (dropdown menu with "CEC"), "Fax:" (text box with "214-555-5551"), and "E-mail:" (dropdown menu). At the bottom of the window are four buttons: "< Previous", "Next >", "Finish", and "Cancel".

New Case

Welcome Case Properties **Additional Information**

Additional Information
Please enter additional information. This information is not required and can be filled any time through the Properties pane.

Investigator name: Agency/Company:
Chuck Easttom CEC

Phone: Fax:
214-555-5555 214-555-5551

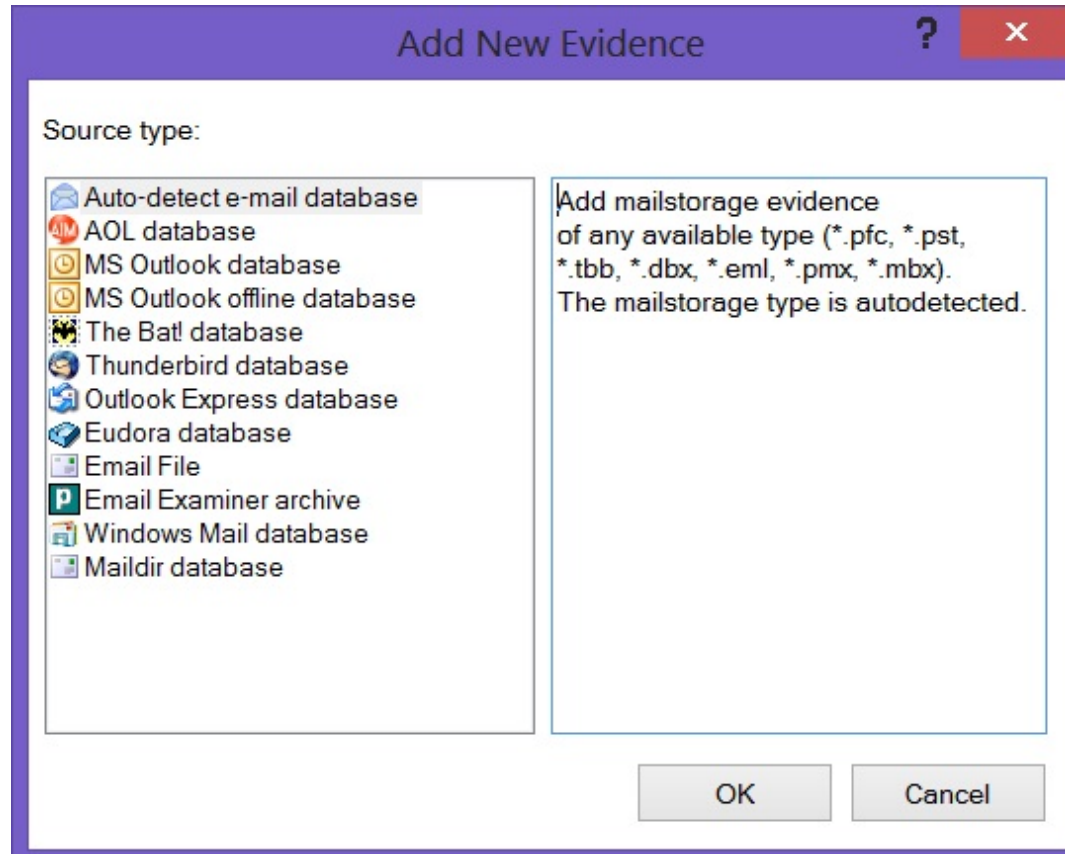
Address: E-mail:

Comments:

< Previous Next > Finish Cancel

Courtesy of Paraben Corporation

Selecting an Email Database



Courtesy of Paraben Corporation

Tracing Email

- Looking at each point through which an email passed and working step by step back to the originating computer

Email Server Forensics

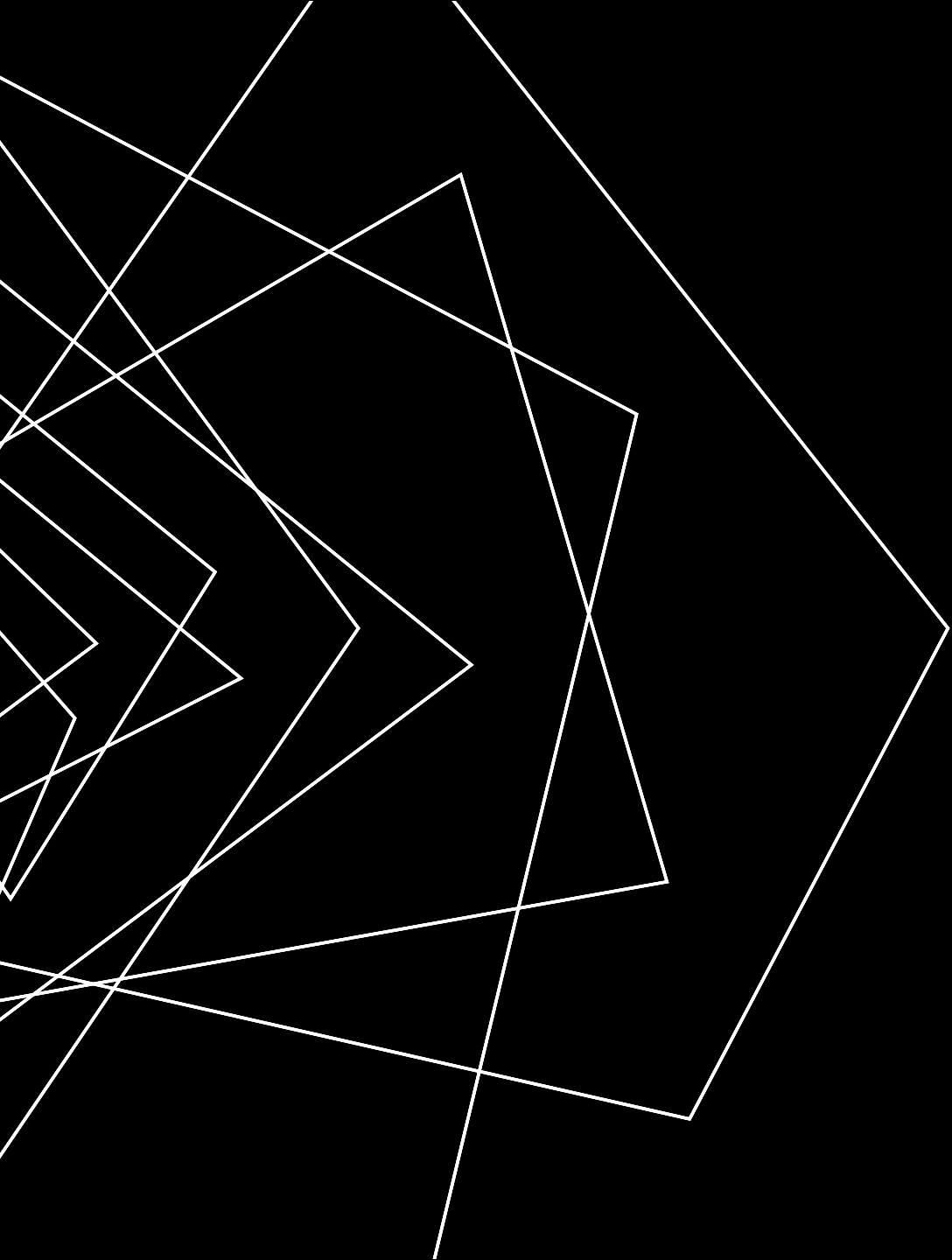
- Examining email servers
 - Searching through deleted emails retained by the server
- Many servers have a retention policy

Email Laws

- The Fourth Amendment to the U.S. Constitution
- The Electronic Communications Privacy Act (ECPA)
- The CAN-SPAM Act
- 18 U.S.C. 2252B
- Communication Assistance for Law Enforcement Act (CALEA)
- Foreign Intelligence Surveillance Act (FISA)
- The USA PATRIOT Act

Summary

- Email clients and servers
- Email headers
- Email tracing
- Email server forensic examination
- Laws related to email investigations



THANK YOU

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ACB1 Room 335